

Counting in Probability

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MATH 241

Lecture #3

Reminders!

- Homework #1 is due on Friday, 9/11 at 11:59pm
 - ▶ Probability problems plus an R problem
- Checkpoint #1 on Thursday (at the end of class)
 - ▶ 1-2 questions, on your own, use of written/printed notes only
- Office hours: today from 3-4:30pm in Rocky 106A

Learning Objectives

- Connect principles of naive definition of probability to a general mathematical definition
- Apply rules of counting to components of probability problems, such as the multiplication rule and sampling

Naive Definition of Probability

Suppose all outcomes of an experiment are equally-likely, then the probability of an event A is given by:

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } S}$$

- i.e. empirical probability

Example

We'll start with a classic probability setting: picking balls from an urn. Suppose we have the following setup:

- 20 balls in urn
- 7 green, 3 blue, 6 red, 4 purple

Suppose we pick one ball from the urn. Then,

- What is the probability of picking a red ball?
- What is the probability of picking a non-red ball?
- What is the probability of picking a red or blue ball?

Limitations of Naive Approach

- S has to be finite
- Equal chance for each item to be pulled out
- Stable (and simple) model
- Anything else?

This motivates our segue into an axiomatic definition of probability, one that will be useful as a foundation for other pieces of probability.

Counting: From a Baby to Now

- One of the first tools we learn as a baby
 - ▶ Base-10
- The principles apply here too, but with the extension of combinations and permutations
- Do these ring a bell?

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$
$$P_{n,k} = \frac{n!}{(n-k)!}$$

Addition Rule

- Choosing object from a objects or choose object from b objects
 - ▶ $a + b$ options

Multiplication Rule

- Choosing object from a objects and another object from b objects
 - ▶ $a \times b$ options

Inclusion-Exclusion Principle

- Choosing object from a objects or choose object from b objects, where two categories have c objects in common
 - ▶ $a + b - c$ options

Combination

- Choose k objects from n unique objects **where order does not matter**

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Permutation

- Choose k objects from n unique objects **where order matters**

$$P_{n,k} = \frac{n!}{(n-k)!}$$

Example #1: Voting

In a random sample of 1000 eligible voters, 500 of them voted in the primary, 640 of them voted in the general election, and 430 of them voted in both elections.

- What is the probability that someone voted in either the primary or general election?

Example #2: A PhD Graduate's Dream

Suppose I had started a bakery instead of staying in academia after my PhD. For a baked good, I want to choose one fruit and two toppings out of the 4 possible fruits and 5 possible toppings.

- How many possible options do I have?

Example #3: Combinations

- How many ways are there to pick 2 students from 3 juniors and 3 seniors?
- How many ways are there to choose 9 players from a bench of 13?
- How many ways are there to choose get a plate of 3 protein and 2 sides from 12 different proteins and 5 different sides?

Example #4: Permutations

- How many ways are there to pick 2 students from 3 juniors and 3 seniors for president and vice president?
- How many ways could you assign 1 gold, 1 silver, and 1 bronze medal to a pool of 12 competing teams?
- How many ways are there for permute the letters in the word FALAFEL?

Sampling

How we sample from some population, space, or pool will dictate the setup of our probability problems:

- Without replacement: I draw one item, and that item is no longer in consideration for a future draw
- With replacement: I draw one item, and that item gets returned for a future draw

Example

Let's return to this question: picking balls from an urn. Suppose we have the following setup:

- 20 balls in urn
- 7 green, 3 blue, 6 red, 4 purple

Suppose we pick one red ball from the urn. Then,

- What is the probability of picking a red ball on the next draw if we sample with replacement? Without replacement?
- Say we complete the second draw, which is a red ball. What is the probability of picking a red ball on the next draw if we sample with replacement? Without replacement?

Table of Outcomes

	Ordered	Not ordered
With replacement	n^k	$\binom{n+k-1}{k}$
W/o replacement	$P_{n,k}$	$\binom{n}{k}$

The Birthday Paradox

Question: At least how many people need to be in a room such that there is a 50% chance that any two people in the room share the same birthday?

The Birthday Paradox

Question: At least how many people need to be in a room such that there is a 50% chance that any two people in the room share the same birthday? Discuss with 2-3 people next to you and do the following:

- Take a slip of paper and make a guess.
- How would you set up the problem? What items would you need to solve?

See You Thursday!

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